

Chemistry 30 | Thermochemical Changes | Lesson 7

Bond Energy and Reactions — Part A: Bonds as Energy Accounting

bond breaking vs bond forming • bond energies • estimating ΔH (approx.)

Big Idea

Bond energy is a practical way to *estimate* how much energy a reaction will absorb or release.

The key idea is simple but easy to mix up: breaking bonds requires energy (endothermic), while forming bonds releases energy (exothermic). A reaction's overall enthalpy change (ΔH) comes from the competition between those two energy changes.

1) Activation energy: why reactions need a “push”

When we say a chemical reaction “releases energy” or “absorbs energy,” we are really talking about changes in the potential energy stored in the electrostatic forces between atoms.

Think of a bond as a stable, low-energy arrangement. Atoms bond because the bonded arrangement is *more stable* than the separated atoms. That stability is why energy must be *added* to pull bonded atoms apart. In other words, *you cannot break a bond and harvest energy from it*. Breaking a bond is always an energy-in step.

So where does the energy come from in an exothermic reaction?

It comes from *forming new bonds* that are, on average, stronger (lower energy) than the bonds that were broken. The system “pays” energy to break old bonds, then “earns” energy back—often more—when new bonds form. The net difference is the reaction's ΔH .

This is why bond energies are useful: they let us estimate ΔH without doing calorimetry. A bond energy (sometimes called an average bond enthalpy) is the average energy required to break *one mole of a specific bond* in the gas phase.

The calculation idea:

- Add up the energy required to break all bonds in the reactants (energy in).
- Add up the energy released when bonds form in the products (energy out).

- Compare them.

We use the relationship:

$$\Delta H \approx \Sigma(\text{E bonds broken}) - \Sigma(\text{E bonds formed})$$

Why the minus sign? Because bond formation releases energy. Subtracting the “energy out” keeps the accounting consistent.

A quick sign check:

If $\Sigma(\text{E broken}) > \Sigma(\text{E formed})$, then ΔH is positive $\rightarrow$ endothermic.

If $\Sigma(\text{E broken}) < \Sigma(\text{E formed})$, then ΔH is negative $\rightarrow$ exothermic.

Worked example (bond-energy estimate):

Estimate ΔH for: $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2 \text{HCl}(\text{g})$

Step 1 — Identify bonds broken (reactants):

- 1 $\times$ H-H
- 1 $\times$ Cl-Cl

Step 2 — Identify bonds formed (products):

- 2 $\times$ H-Cl

Step 3 — Substitute typical bond energies (kJ/mol):

H-H $\approx$ 436 Cl-Cl $\approx$ 243 H-Cl $\approx$ 431

Step 4 — Calculate:

Energy in = 436 + 243 = 679 kJ

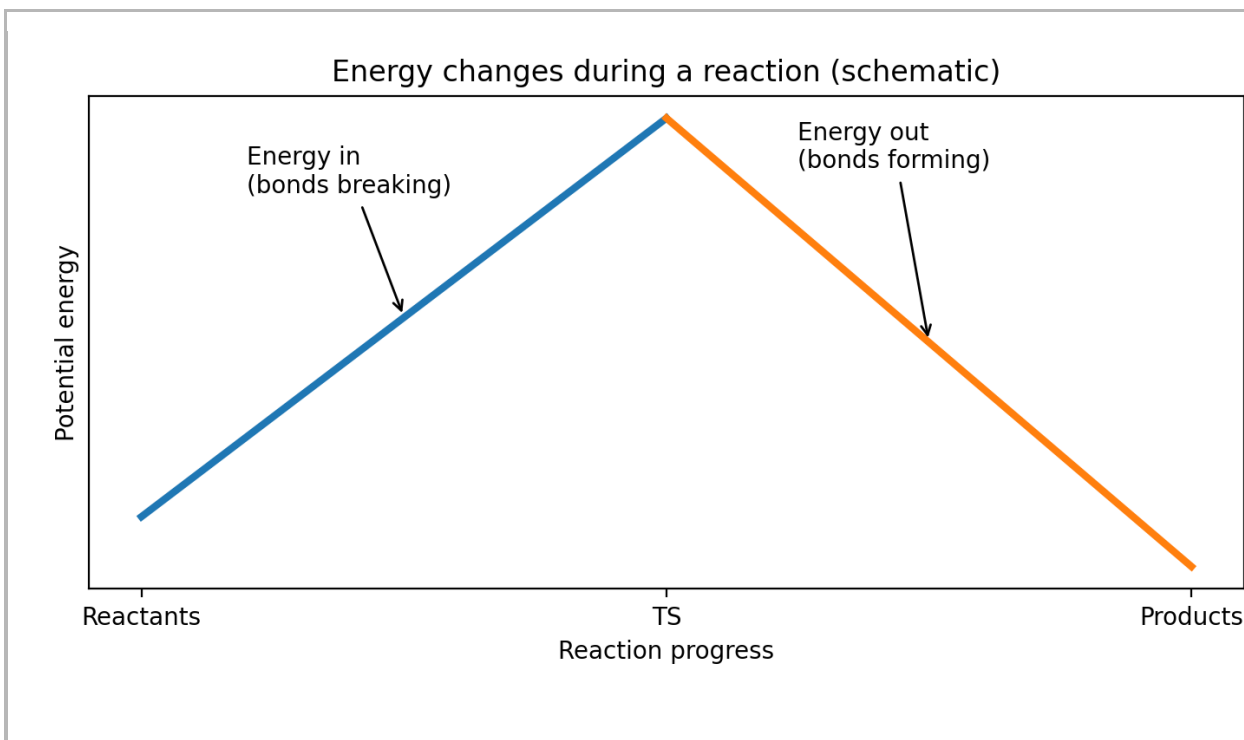
Energy out = 2(431) = 862 kJ

$\Delta H \approx 679 - 862 = -183$ kJ

Interpretation: ΔH is negative, so the reaction is exothermic. The products form stronger total bonding than the reactants started with.

Important realism check (why this is an estimate):

Bond energies are *averages*. The energy of a C-H bond in methane is not identical to a C-H bond in ethane. Different molecular environments change the exact bond strength. That is why ΔH from bond energies is approximate, while ΔH calculated from formation enthalpies can be much more accurate.



[Instructional narrative text goes here]

Key Vocabulary

Bond energy (average bond enthalpy): the average energy required to break one mole of a specific bond in gaseous molecules.

Bond dissociation energy: the energy required to break a particular bond in a particular molecule (more specific than an average bond energy).

ΔH (enthalpy change): the net energy absorbed or released by the system at constant pressure.

Estimate: a calculated value that is close but not exact because it relies on averaged or simplified data.

Common Error

Saying “breaking bonds releases energy.” This is backwards.

If you remember only one sentence: ****breaking bonds costs energy; forming bonds pays energy back.****

Also watch the counting: you must multiply each bond energy by the number of those bonds that actually break or form according to the balanced equation.

Diploma Tip

On diploma-style questions, use the structure:

- 1) Write the balanced chemical equation.
- 2) List bonds broken (reactants) and count them.
- 3) List bonds formed (products) and count them.
- 4) Apply: $\Delta H \approx \Sigma_{\text{broken}} - \Sigma_{\text{formed}}$.

Then add a one-sentence interpretation: exothermic/endothermic *and* what that means for energy flow (system $\rightarrow$ surroundings or surroundings $\rightarrow$ system).

Pause & Think

A student says: “If bond formation releases energy, then strong bonds must *contain* lots of energy.”

Explain what is wrong with that statement using the idea of *stability* (lower energy) versus “stored energy.”

Then answer: Would breaking a very strong bond require more or less energy than breaking a weak bond? Why?

Part B — Estimating ΔH Using Bond Energies

Big Idea

Bond energies let us reason about reaction energy changes by focusing on what is happening to bonds. Breaking bonds always requires an input of energy, while forming bonds releases energy. If we can count which bonds are broken and which new bonds are formed, we can estimate the reaction enthalpy change (ΔH). This method is fast, logical, and especially useful when formation-enthalpy data are not available—but it produces an estimate because bond energies are average values.

1) The Bond-Energy Method: Estimating ΔH 1) Activation energy: why reactions need a “push”

In earlier lessons, you learned that ΔH tells us whether energy is released to the surroundings (exothermic, $\Delta H < 0$) or absorbed from the surroundings (endothermic, $\Delta H > 0$). Bond energies give us a different way to get at the same idea. Instead of starting with tabulated ΔH°_f values, we start by looking at the bonds that must be broken and the bonds that must be formed in order for the reaction to happen.

Here is the key reasoning: to break a chemical bond, you must supply energy to pull atoms apart against the attractive forces holding them together. That energy input is always positive. When a new bond forms, atoms fall into a lower-energy, more stable arrangement, and energy is released. That energy output is always negative from the system's point of view.

Because a reaction is essentially a ‘trade’—old bonds are broken and new bonds are formed—the overall energy change depends on which side wins: if forming the new bonds releases more energy than breaking the old bonds requires, the reaction is exothermic. If breaking bonds requires more energy than forming bonds releases, the reaction is endothermic.

We capture that idea with one extremely useful relationship:

$$\Delta H_{\text{rxn}} \approx \Sigma E(\text{bonds broken}) - \Sigma E(\text{bonds formed})$$

Notice what the formula is doing. We first add up the energy needed to break every bond that disappears (positive contributions). Then we subtract the total energy released when the new

bonds appear. The subtraction is not a 'sign trick'—it reflects the fact that bond formation releases energy.

A practical workflow that keeps students out of trouble looks like this:

Step 1: Write the balanced chemical equation. The coefficients matter because they change how many molecules react, and therefore how many bonds are involved.

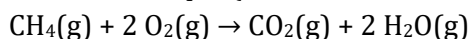
Step 2: Sketch simple structural formulas (or at least list the types of bonds) for each reactant and product. This step prevents counting the wrong bonds.

Step 3: Count bonds broken (reactant side) and multiply by bond energies.

Step 4: Count bonds formed (product side) and multiply by bond energies.

Step 5: Apply $\Delta H \approx \Sigma(\text{broken}) - \Sigma(\text{formed})$, then interpret the sign and the size.

Worked Example (methane combustion): Consider the combustion reaction



To estimate ΔH using average bond energies, we identify bonds broken and formed.

Bonds broken (reactants):

- CH_4 has 4 C–H bonds $\rightarrow 4 \times E(\text{C–H})$
- 2 O_2 has 2 O=O bonds $\rightarrow 2 \times E(\text{O=O})$

Bonds formed (products):

- CO_2 has 2 C=O (double) bonds $\rightarrow 2 \times E(\text{C=O})$
- 2 H_2O has 4 O–H bonds $\rightarrow 4 \times E(\text{O–H})$

Using typical average bond energies (kJ/mol): C–H 413, O=O 498, C=O (in CO_2) 799, O–H 463

Energy to break bonds:

$$4(413) + 2(498) = 1652 + 996 = 2648 \text{ kJ}$$

Energy released forming bonds:

$$2(799) + 4(463) = 1598 + 1852 = 3450 \text{ kJ}$$

Estimated reaction enthalpy:

$$\Delta H \approx 2648 - 3450 = -802 \text{ kJ}$$

Interpretation: The negative sign tells us the system loses energy overall—this is an exothermic reaction. Also notice the magnitude: hundreds of kilojoules per 'reaction as written' is consistent with combustion being a major energy source.

Why is this only an estimate? Average bond energies are measured for many different molecules and environments, usually in the gas phase. In real reactions, the exact energy of a

bond depends on the molecule it is in, the surrounding atoms, and the physical state. So your calculated ΔH from bond energies should be close to the real value, but not identical.

Key Vocabulary Key Vocabulary

Bond energy (bond dissociation energy): the energy required to break one mole of a specific bond in gaseous molecules. Stronger bonds have larger bond energies.

Average bond energy: a bond energy value averaged over many different molecules. This is why bond-energy ΔH calculations are approximate.

Bonds broken / bonds formed: the 'accounting list' for a reaction. In calculations, broken bonds contribute positive energy; formed bonds contribute negative energy (released).

Reaction profile (energy diagram): a graph of potential energy versus reaction progress. Activation energy (E_a) is the barrier height; ΔH is the difference between products and reactants.

Common Error

Students often get the right idea but the wrong number because of bond counting. The two most common mistakes are (1) forgetting to multiply by coefficients (for example, 2 O_2 means two $O=O$ bonds), and (2) misidentifying the bonds in the products (for example, CO_2 has two $C=O$ double bonds, not one). A good habit is to list bonds explicitly before you ever touch your calculator.

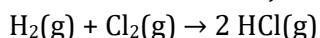
Diploma Tip

When a question gives you a bond-energy table, it is testing reasoning, not memorization. Show your 'bond inventory' clearly (bonds broken and bonds formed) and include coefficient multipliers. Then state what the sign means in words: ' ΔH is negative, so energy is released to the surroundings.' If your estimate is not close to a

known ΔH from formation enthalpies, do not panic—explain that bond-energy values are averages.

Pause & Think

Without doing a full calculation, predict whether the reaction below is likely exothermic or endothermic, and justify your reasoning using bond ideas:



Hint: Which bonds are broken, and which bonds are formed? Which do you expect to be stronger?

Chemistry 30 — Unit II | Lesson 7

Bond Energy and Reactions

PART C — Using Bond Energies to Estimate Reaction ΔH

Big Idea

Bond energies let us estimate how much energy a reaction transfers by comparing energy absorbed to break reactant bonds with energy released when product bonds form. The method is powerful for reasoning—but the answer is always an approximation because bond energies are averages.

1) The bond-energy method: what you are actually calculating

A bond energy is the amount of energy required to break a specific covalent bond. If you break a bond, you must put energy in. If you form a bond, the opposite happens: energy is released as the particles settle into a lower-energy arrangement.

That is why the bond-energy method works like a “balance sheet.” You add up the energy needed to break the reactant bonds, then you subtract the energy released when product bonds form. What is left over is the net energy change for the system (ΔH).

The key idea is that ΔH is not about one bond; it is about the overall change from reactants to products. So the method only works if you count every bond that is broken and every bond that is formed in the balanced reaction.

In symbols, we write: $\Delta H \approx \Sigma E(\text{bonds broken}) - \Sigma E(\text{bonds formed})$. Read it as: “energy in minus energy out.”

Key Vocabulary

Bond energy (bond enthalpy): the energy required to break one mole of a specific covalent bond (values are typically averages).

Average bond energy: a single “typical” value for a bond type (e.g., C–H) taken across many different molecules and environments.

Bond-energy estimate of reaction enthalpy: $\Delta H \approx \Sigma E(\text{bonds broken}) - \Sigma E(\text{bonds formed})$.

Estimate: an answer intended to be reasonable and defensible, not exact to the last digit.

Common Error

Using the formula backwards. If you write $\Sigma(\text{formed}) - \Sigma(\text{broken})$, your sign will be reversed.

Counting coefficients instead of bonds. Coefficients tell you moles of molecules; you still must count bonds per molecule and then multiply.

Forgetting multiple bonds. A double bond (C=C) has its own bond energy and is not counted as a single bond.

Applying bond-energy tables to ionic lattices or network solids. This method is for covalent bonds in molecular substances.

Diploma Tip

Before you calculate, draw or visualize the structural formula for each reactant and product. Your job is to count bond types, not just read formulas.

Set up two clean columns: "Bonds Broken" and "Bonds Formed." Put the number of each bond type beside it, then multiply by the bond energy.

Keep units consistent: bond energies are usually in kJ/mol of bonds. Your final ΔH is kJ per mole of reaction as written.

Worked Example

Worked Example (bond-energy estimate)

Estimate ΔH for: $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$

Use the following average bond energies (kJ/mol): C-H 413, O=O 498, C=O (in CO_2) 799, O-H 463.

Step-by-step:

1) Bonds broken (reactants): CH_4 has 4 C-H; 2O_2 has 2 O=O.

$$\text{Energy in} = 4(413) + 2(498) = 1652 + 996 = 2648 \text{ kJ}$$

2) Bonds formed (products): CO_2 has 2 C=O; $2\text{H}_2\text{O}$ has 4 O-H.

$$\text{Energy out} = 2(799) + 4(463) = 1598 + 1852 = 3450 \text{ kJ}$$

3) $\Delta H \approx \Sigma \text{broken} - \Sigma \text{formed} = 2648 - 3450 = -802 \text{ kJ}$

Conclusion: $\Delta H \approx -802 \text{ kJ}$ (exothermic). Negative ΔH means the system releases energy overall.

Pause & Think

Without calculating, predict whether the combustion of methane should be exothermic or endothermic. What evidence (bond types formed vs broken) supports your prediction?

2) Why bond-energy ΔH values are estimates (and how to use them responsibly)

A bond-energy table does not give the exact energy of the C–H bond in your specific molecule. Instead, it gives an average taken from many molecules. The real bond energy depends on the surrounding atoms, bond polarity, and molecular geometry.

Bond energies are also commonly defined for the gas phase. If your reactants or products are liquids or solids, extra energy changes are involved (intermolecular forces, phase changes) that a simple bond-energy calculation does not include.

That is why you should treat the result as a reasoned estimate. It is excellent for deciding the sign of ΔH and for getting a realistic magnitude, but it will not usually match a data-book value perfectly.

✓ Diploma Tip

When the question says “estimate using bond energies,” you earn marks for a correct method: correct bond counting, correct substitution, and correct sign.

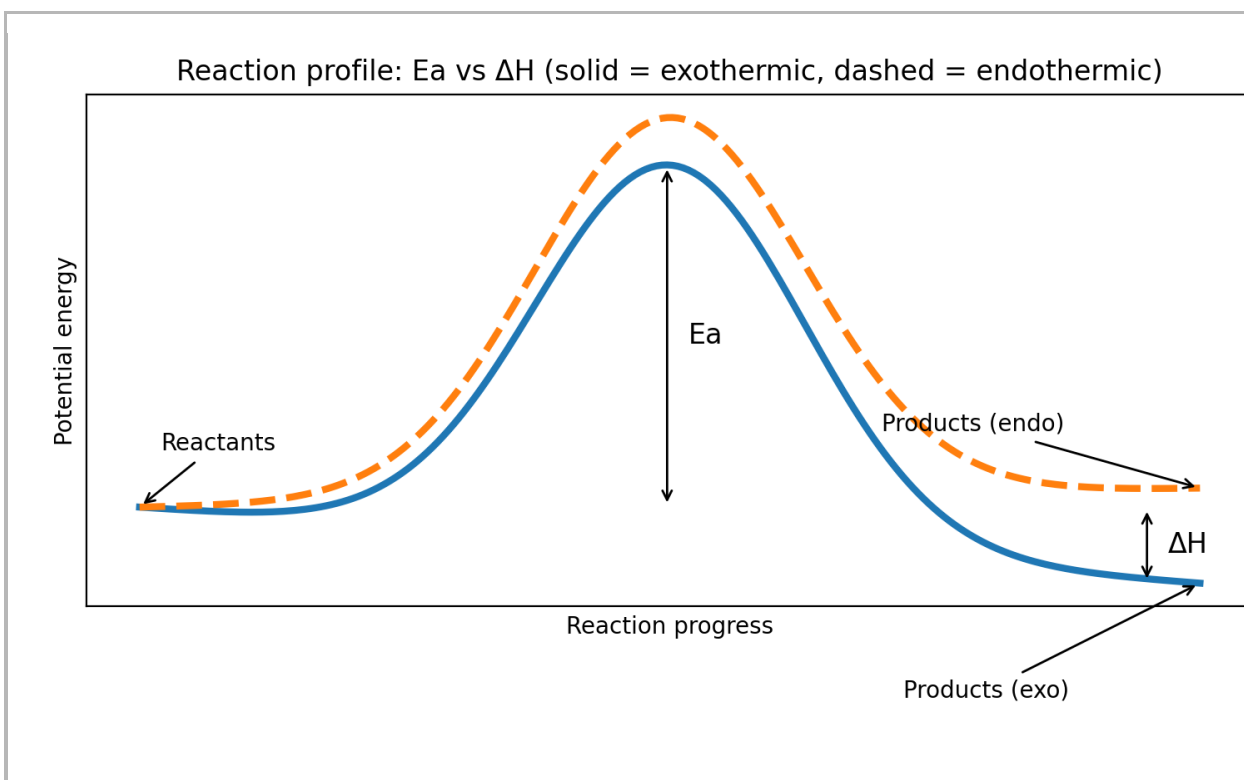
If your answer has the wrong sign, check (1) the formula order, and (2) whether you accidentally swapped “broken” and “formed.”

 3) Bond energies, reaction profiles, and activation energy (connecting to Lesson 6)

Energy diagrams (reaction profiles) are a visual model for the energy bookkeeping you are doing. The curve rises to a peak because reaching the transition state requires partially breaking bonds before the strongest new bonds are fully formed.

Activation energy (E_a) is the height of the barrier from reactants to the peak. ΔH is the difference between products and reactants. They answer different questions: E_a predicts how hard it is to start; ΔH predicts where the energy ends up.

A catalyst lowers E_a by providing an alternate pathway. It does not change ΔH because the initial and final states are unchanged—so the energy ‘start’ and ‘finish’ levels stay the same.



⚡ Bond Energy and Reactions — Part D

✂ Energy diagrams • 🔥 Activation energy • 🧠 Evidence-based exo/endo reasoning

🌱 Big Idea

Energy diagrams are not “art”—they are a picture of energy bookkeeping. They show (1) how much energy is required to get the reaction started (activation energy, E_a) and (2) whether the system finishes at a higher or lower energy than it began (ΔH). Bond energies help you justify the shape and direction of these diagrams: breaking bonds absorbs energy; forming bonds releases energy.

🔗 1) Reaction profiles: what the curve is really showing

An energy (reaction profile) diagram plots the potential energy of the reacting system versus reaction progress. It is a “before → during → after” energy story.

At the left, the reactants sit at a starting potential energy. As the reaction begins, existing bonds

must be stretched and partially broken. That requires an energy input, so the curve rises. The highest point is the transition state (sometimes called the activated complex). The energy gap between the reactants and this peak is the activation energy, E_a .

After the peak, new bonds form as products assemble. Bond formation releases energy, so the curve falls. Where the curve ends (products) compared with where it began (reactants) tells you the enthalpy change, ΔH .

Two ideas keep students from getting lost. First, E_a is about difficulty (rate), not overall energy change. Second, ΔH is about the net energy difference between start and finish—not the height of the hill.

Key Vocabulary

Activation energy (E_a): the minimum energy the reacting particles must gain so a successful collision can reach the transition state.

Transition state / activated complex: the short-lived, highest-energy arrangement of atoms where old bonds are partly broken and new bonds are partly formed.

Reaction coordinate / reaction progress: a bookkeeping axis that represents “how far” the reaction has proceeded; it is not time.

ΔH (enthalpy change): the energy change of the system from reactants to products under constant pressure; $\Delta H < 0$ is exothermic, $\Delta H > 0$ is endothermic.

Common Error

Students often point at the peak and say, “That’s ΔH .” It isn’t.

ΔH is the vertical difference between the reactant level and the product level. The peak is used for activation energy.

A second frequent mistake happens when reversing a diagram. For the reverse reaction, the activation energy is measured from the products up to the same peak—not from the reactants.

✓ Diploma Tip

When a question asks you to label a diagram, treat it like a checklist.

First, label reactants and products (left and right). Next, draw and label ΔH as a vertical arrow from reactants to products. Then draw and label E_a (forward) as a vertical arrow from reactants to the peak. If asked, draw and label E_a (reverse) from products to the peak.

If there are multiple peaks (a multi-step mechanism), the overall ΔH is still start $\rightarrow$ finish, but the forward activation energy is the difference between reactants and the highest peak.

✖ Worked Example — reading E_a and ΔH from an energy diagram

A reaction has the following energy levels (all values in kJ): reactants = 50, peak = 120, products = 20.

Step 1: Find ΔH .

$\Delta H = E(\text{products}) - E(\text{reactants}) = 20 - 50 = -30 \text{ kJ}$. Because ΔH is negative, the reaction is exothermic: the system ends at a lower energy than it started.

Step 2: Find the forward activation energy.

$E_a(\text{forward}) = E(\text{peak}) - E(\text{reactants}) = 120 - 50 = 70 \text{ kJ}$.

Step 3: Find the reverse activation energy.

$E_a(\text{reverse}) = E(\text{peak}) - E(\text{products}) = 120 - 20 = 100 \text{ kJ}$.

Step 4: Connect to bond-energy reasoning.

The uphill portion corresponds to energy absorbed to weaken/break bonds. The downhill portion corresponds to energy released as new bonds form. The net drop of 30 kJ means bond formation releases 30 kJ more than bond breaking absorbs.

⏸ Pause & Think

1) If a catalyst is added, which labeled quantity on the diagram changes: ΔH , E_a , or both? Explain.

2) On an exothermic diagram, which is larger: E_a (forward) or E_a (reverse)? Why?

3) A student says, "If a reaction is exothermic, it must be fast." What would you say back?

2) Using evidence to decide whether a reaction is exothermic or endothermic

In Chemistry 30, you are often asked to justify exothermic vs endothermic using evidence—not just a guess.

Start by naming the system: the reacting chemicals. Then decide what happens to energy in that system. If the system releases energy to the surroundings, the surroundings warm up and ΔH is negative. If the system absorbs energy from the surroundings, the surroundings cool down and ΔH is positive.

Bond energies give you a second, independent line of evidence. Ask: do you expect the bonds in the products to be stronger (lower energy) or weaker (higher energy) than those in the reactants? Forming stronger bonds releases more energy, pushing ΔH negative. Forming weaker bonds releases less energy, pushing ΔH positive.

Important: temperature change is observed in the surroundings. The sign of ΔH is about the system. If you keep those roles straight, the reasoning becomes consistent.

Diploma Tip — the three-way consistency check

When you answer exothermic/endothermic questions, make sure these three statements agree.

(A) Sign: $\Delta H < 0$ (exo) or $\Delta H > 0$ (endo).

(B) Energy flow: "Energy is released by the system" (exo) or "Energy is absorbed by the system" (endo).

(C) Diagram: products lower than reactants (exo) or products higher than reactants (endo).

If one statement disagrees, stop and fix the logic before moving on.

3) Catalysts, alternative pathways, and why ΔH does not change

A catalyst provides an alternative pathway with a lower activation energy. On an energy diagram, the peak becomes lower. However, the reactants and products are the same substances, so their energy levels do not change. Therefore the vertical difference between them— ΔH —stays the same.

This connects directly to Hess' Law: ΔH depends only on the initial and final states, not on the pathway. A catalyst changes the pathway (how the reaction happens), not the endpoints (what you start with and end with). So catalysts can make reactions faster and safer without changing the overall energy released or absorbed.

Misconception check: a lower peak does not mean less energy released. The peak is the barrier; the vertical difference between reactants and products is the net energy change.

Part E — Why Bond-Energy ΔH Values Are Estimates (and how to use them well)

Big Idea

Bond energies let us *estimate* reaction enthalpy by comparing the energy required to break bonds with the energy released when new bonds form. But bond energies are *averages*, so our ΔH value is a best-reasoned approximation—not a lab-measured exactness.

Mini-Lesson 1 — What a “bond energy” number really means

When you look up a bond energy in a data table, it is tempting to treat it like a precise property—like the boiling point of ethanol or the density of aluminum. Bond energies are not like that. Most tables list *mean bond energies* (also called average bond enthalpies). That word *mean* matters.

A C–H bond in methane is not exactly the same as a C–H bond in ethanol. The atoms are different, but the *environment* around the bond is also different: nearby atoms pull electron density, bond angles change, and some bonds are strengthened or weakened by resonance, polarity, or bond order. Because of this, the energy required to break “a C–H bond” is not a single universal value.

To make bond energies useful, chemists measure (or calculate) many related bonds and report

an **average value** that works well *on average* across a wide range of molecules. Most bond energy tables are also based on the **gas phase** (particles far apart), because that is the cleanest situation for comparing bond-breaking energies. In liquids and solids, extra attractions between particles can change the energy bookkeeping.

Here is the practical takeaway: bond energies are a powerful *model*. They are designed to help you reason about energy changes and to make reasonable predictions—especially when you do not have formation enthalpies for every compound you care about. But the price you pay is that your ΔH will be an estimate.

An energy (reaction profile) diagram plots the potential energy of the reacting system versus reaction progress. It is a “before $\rightarrow$ during $\rightarrow$ after” energy story.

At the left, the reactants sit at a starting potential energy. As the reaction begins, existing bonds must be stretched and partially broken. That requires an energy input, so the curve rises. The highest point is the transition state (sometimes called the activated complex). The energy gap between the reactants and this peak is the activation energy, E_a .

After the peak, new bonds form as products assemble. Bond formation releases energy, so the curve falls. Where the curve ends (products) compared with where it began (reactants) tells you the enthalpy change, ΔH .

Two ideas keep students from getting lost. First, E_a is about difficulty (rate), not overall energy change. Second, ΔH is about the net energy difference between start and finish—not the height of the hill.



Key Vocabulary (use these precisely)

- **Mean bond energy / average bond enthalpy** — an average value for a bond type across many molecules
- **Bond dissociation energy (BDE)** — energy required to break one mole of a specific bond in a specific molecule (more precise than a mean value)
- **Energy absorbed** — energy required to *break* bonds (input)
- **Energy released** — energy given off when bonds *form* (output)
- **Estimate** — a value you justify with evidence and reasoning, not a guaranteed exact result
- **ΔH vs E_a** — overall energy change vs the energy barrier to start the reaction

Activation energy (E_a): the minimum energy the reacting particles must gain so a successful collision can reach the transition state.

Transition state / activated complex: the short-lived, highest-energy arrangement of atoms where old bonds are partly broken and new bonds are partly formed.

Reaction coordinate / reaction progress: a bookkeeping axis that represents “how far” the reaction has proceeded; it is not time.

ΔH (enthalpy change): the energy change of the system from reactants to products under constant pressure; $\Delta H < 0$ is exothermic, $\Delta H > 0$ is endothermic.

⚠ Common Error (and why it happens)

Students often write the bond-energy equation correctly, but then *count the wrong bonds*.

Remember the logic: you only include bonds that **change**. Bonds that appear unchanged on both sides are not “broken” or “formed” overall.

A reliable habit is to:

(1) draw (or visualize) reactant and product structures, (2) list bonds that must be broken to separate reactants into atoms, and (3) list the new bonds that must form to build the products.

Another common slip is mixing up the sign and accidentally computing “energy released – energy absorbed.” Keep repeating this sentence to yourself: **Breaking costs energy. Forming pays energy back.**

Students often point at the peak and say, “That’s ΔH .” It isn’t.

ΔH is the vertical difference between the reactant level and the product level. The peak is used for activation energy.

A second frequent mistake happens when reversing a diagram. For the reverse reaction, the activation energy is measured from the products up to the same peak—not from the reactants.

✅ Diploma Tip (what markers reward)

When a question says **estimate using bond energies**, it is signalling that an approximate answer is acceptable. Markers look for clear reasoning more than perfect precision.

Always show the structure of your calculation:

$$\Delta H \approx \Sigma(\text{E bonds broken}) - \Sigma(\text{E bonds formed})$$

Then write one sentence interpreting the sign:

- $\Delta H < 0 \rightarrow$ exothermic (system releases energy)
- $\Delta H > 0 \rightarrow$ endothermic (system absorbs energy)

Finally, state the unit clearly: ****kJ per mole of reaction as written****.

When a question asks you to label a diagram, treat it like a checklist.

First, label reactants and products (left and right). Next, draw and label ΔH as a vertical arrow from reactants to products. Then draw and label E_a (forward) as a vertical arrow from reactants to the peak. If asked, draw and label E_a (reverse) from products to the peak.

If there are multiple peaks (a multi-step mechanism), the overall ΔH is still start $\rightarrow$ finish, but the forward activation energy is the difference between reactants and the highest peak.



Worked Example — Estimating ΔH with bond energies

Estimate ΔH for the reaction: $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$

Step 1: Identify bonds broken (energy absorbed)

- 1 H-H bond
- 1 Cl-Cl bond

Step 2: Identify bonds formed (energy released)

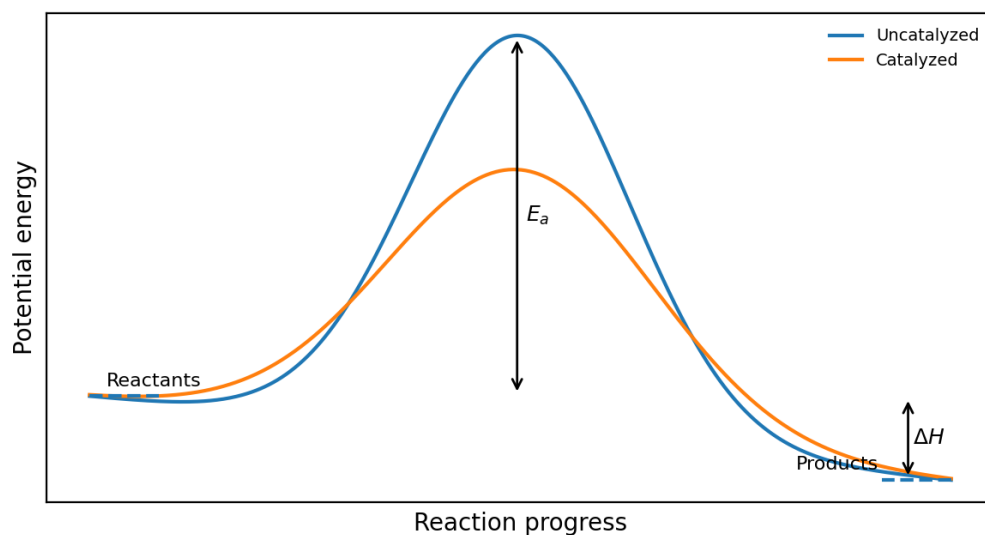
- 2 H-Cl bonds

Step 3: Apply the bond-energy model

$$\Delta H \approx [E(\text{H-H}) + E(\text{Cl-Cl})] - [2 \times E(\text{H-Cl})]$$

Step 4: Interpret your result

If the energy released by forming two H-Cl bonds is larger than the energy needed to break H-H and Cl-Cl, ΔH will be negative. That means the system releases energy and the surroundings warm up.



A reaction has the following energy levels (all values in kJ): reactants = 50, peak = 120, products = 20.

Step 1: Find ΔH .

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Step 4: Connect to bond-energy reasoning.

The uphill portion corresponds to energy absorbed to weaken/break bonds. The downhill portion corresponds to energy released as new bonds form. The net drop of 30 kJ means bond formation releases 30 kJ more than bond breaking absorbs.

✿ Pause & Think

Without calculating, decide whether this reaction is **likely** exothermic or endothermic:
 $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$

Use evidence: Which strong bonds are formed in the products? Which bonds must be

broken in the reactants? Write a one-sentence prediction *and* the reason for your prediction.

1) If a catalyst is added, which labeled quantity on the diagram changes: ΔH , E_a , or both? Explain.

2) On an exothermic diagram, which is larger: E_a (forward) or E_a (reverse)? Why?

3) A student says, "If a reaction is exothermic, it must be fast." What would you say back?



Mini-Lesson 2 — Why your estimate may not match a tabulated ΔH

Suppose you calculate ΔH using bond energies and then compare it to a ΔH value from formation enthalpies or from calorimetry. It is normal for the numbers to differ—sometimes by tens of kJ/mol. That difference is not "you doing it wrong." It is the model showing its limits.

There are three main reasons:

First, **averaging**: bond energies are mean values. The actual bond energies in your specific molecules may be higher or lower than the average.

Second, **state and environment**: bond-energy tables assume gas-phase bond breaking. Real reactions may involve liquids, solids, or solutions, where particle attractions and phase changes shift the energy accounting.

Third, **what ΔH measures**: the best ΔH values come from a state-function method (formation enthalpies or direct measurement). Those values account for the real starting and ending states of the substances. Bond energies approximate the same idea by pretending that all bonds of a type behave similarly.

Practical conclusion: bond energies are excellent for *direction* (exo vs endo), for *relative comparisons*, and for reasonable estimates—but if your course provides formation enthalpies, those give the more reliable ΔH .

In Chemistry 30, you are often asked to justify exothermic vs endothermic using evidence—not just a guess.

Start by naming the system: the reacting chemicals. Then decide what happens to energy in that system. If the system releases energy to the surroundings, the surroundings warm up and ΔH is negative. If the system absorbs energy from the surroundings, the surroundings cool down and ΔH is positive.

Bond energies give you a second, independent line of evidence. Ask: do you expect the bonds in the products to be stronger (lower energy) or weaker (higher energy) than those in the reactants? Forming stronger bonds releases more energy, pushing ΔH negative. Forming weaker bonds releases less energy, pushing ΔH positive.

Important: temperature change is observed in the surroundings. The sign of ΔH is about the system. If you keep those roles straight, the reasoning becomes consistent.

✓ Diploma Tip — Justifying “exo” or “endo” with evidence

In written-response questions, avoid answers like “It’s exothermic because ΔH is negative” if you never explained where the sign came from.

A strong justification sounds like this:

“More energy is released when strong bonds form in the products than is required to break the bonds in the reactants. Therefore, ΔH is negative and the reaction is exothermic.”

Notice what this does: it names the evidence (bond breaking vs bond forming) and then connects that evidence to the sign of ΔH .

When you answer exothermic/endothermic questions, make sure these three statements agree.

(A) Sign: $\Delta H < 0$ (exo) or $\Delta H > 0$ (endo).

(B) Energy flow: “Energy is released by the system” (exo) or “Energy is absorbed by the system” (endo).

(C) Diagram: products lower than reactants (exo) or products higher than reactants (endo).

If one statement disagrees, stop and fix the logic before moving on.



Mini-Lesson 3 — Connecting bond energy reasoning to reaction profiles and activation energy

Bond-energy calculations help you estimate ΔH , the vertical difference between reactants and products on an energy diagram. Activation energy, E_a , is different: it is the size of the “hill” that must be climbed to reach the transition state.

Here is the key distinction that students often blur:

- ΔH is about the destination. It depends only on the energy of reactants and products.

- **E_a is about the route.** It depends on what bonds must begin breaking and forming in the transition state.

Bond strengths still matter for E_a , because breaking strong bonds early can raise the barrier. But you cannot compute E_a with the same simple bond-energy subtraction used for ΔH . That is why catalysts can lower E_a without changing ΔH : they change the pathway (the hill), not the start and end levels.

If you keep this separation clear, reaction profile questions become much easier: ΔH answers “Where does the energy end up?” and E_a answers “How hard is it to get the reaction started?”

A catalyst provides an alternative pathway with a lower activation energy. On an energy diagram, the peak becomes lower. However, the reactants and products are the same substances, so their energy levels do not change. Therefore the vertical difference between them— ΔH —stays the same.

This connects directly to Hess’ Law: ΔH depends only on the initial and final states, not on the pathway. A catalyst changes the pathway (how the reaction happens), not the endpoints (what you start with and end with). So catalysts can make reactions faster and safer without changing the overall energy released or absorbed.

Misconception check: a lower peak does not mean less energy released. The peak is the barrier; the vertical difference between reactants and products is the net energy change.

Lesson 7 — Part F

Guided Practice + Diploma-Style Questions + Full Worked Solutions

★ Big Idea

Bond energies let us *estimate* the enthalpy change (ΔH) of a reaction by treating bonds like energy “accounts”: breaking bonds costs energy; forming bonds pays energy back. When we compare the total energy in and out, we can justify whether a reaction is likely exothermic or endothermic—and connect that reasoning to reaction profiles and activation energy.

STUDENT PRACTICE — Questions Only

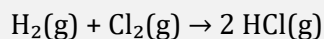
How to Use This Practice

- 1) Read the reaction carefully and make sure it is balanced before you start.
- 2) Identify which bonds are *broken* (reactants) and which are *formed* (products).
- 3) Use the bond-energy relationship: $\Delta H \approx \Sigma E(\text{bonds broken}) - \Sigma E(\text{bonds formed})$.
- 4) Decide whether your sign makes physical sense: exothermic ($\Delta H < 0$) releases energy; endothermic ($\Delta H > 0$) absorbs energy.
- 5) When you compare two pathways (catalyzed vs uncatalyzed), remember: catalysts change E_a , not ΔH .

Guided Practice — Bond Energies, ΔH Estimates, and Energy Profiles

Question 1 — Estimating ΔH from Bond Energies

Use the bond energies below to estimate ΔH for the reaction, then classify it as exothermic or endothermic:



Bond energies (kJ/mol): H-H = 436, Cl-Cl = 243, H-Cl = 431

- a) Identify the bonds broken and the bonds formed.
- b) Calculate ΔH using $\Delta H \approx \Sigma(\text{broken}) - \Sigma(\text{formed})$.
- c) Interpret your answer in terms of energy flow (system vs surroundings).

Question 2 — Connecting Bond Energies to a Reaction Profile

Consider an exothermic reaction represented by a reaction-profile (potential energy vs reaction progress).

- a) Explain, in words, what the “hill” (peak) represents at the particle level.
- b) On a typical exothermic profile, where would you show E_a (forward) and ΔH ? What would be the sign of ΔH ?
- c) If the bonds formed in the products are *much stronger* than the bonds broken in the reactants, what happens to the relative height of the products compared to the reactants on the profile? Explain.

Questions 3–4 — Concept Check (Explain Your Reasoning)

3) A student uses bond energies to calculate ΔH and gets a value that does not match the value from a table of ΔH°_f . Give two reasons why the bond-energy result is an *estimate*, not an exact value.

4) A catalyst is added and the reaction becomes faster. Explain why this does *not* mean the reaction releases more energy. Use the ideas of E_a and ΔH in your explanation.

 Multiple Choice — Diploma-Style Application (Questions 5–12)

5) Which statement best explains why *breaking bonds* is always an energy-absorbing process?

- A. Breaking a bond releases energy stored in the bond.
- B. Breaking a bond requires energy to overcome the attraction between atoms.
- C. Breaking a bond increases stability by lowering potential energy.
- D. Breaking a bond happens spontaneously whenever temperature increases.

6) A student estimates ΔH for a reaction using bond energies and finds $\Delta H \approx -250$ kJ. The balanced equation is then doubled (all coefficients $\times 2$). What is the new estimated ΔH ?

- A. -125 kJ
- B. -250 kJ
- C. -500 kJ
- D. $+500$ kJ

7) A reaction is observed to be strongly exothermic. Which bond-energy statement must be true?

- A. Total energy required to break reactant bonds is greater than total energy released when product bonds form.
- B. Total energy released when product bonds form is greater than total energy required to break reactant bonds.
- C. Activation energy must be negative.
- D. The products must have weaker bonds than the reactants.

8) Two reaction profiles are shown for the same reaction: one catalyzed, one uncatalyzed. Which change is expected when a catalyst is used?

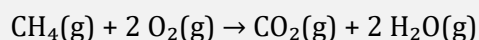
- A. E_a decreases; ΔH becomes more negative.
- B. E_a increases; ΔH stays the same.
- C. E_a decreases; ΔH stays the same.
- D. E_a stays the same; ΔH becomes zero.

DIPLOMA-STYLE QUESTIONS — Mixed Response (MC + Written)

9) A student compares two reactions using bond-energy reasoning. Reaction X breaks several strong bonds and forms several weak bonds. Reaction Y breaks weak bonds and forms strong bonds. Which is the best prediction?

- A. Reaction X is more likely exothermic than Reaction Y.
- B. Reaction Y is more likely exothermic than Reaction X.
- C. Both must be endothermic.
- D. Both must be exothermic.

10) For the reaction below, which bonds are counted in $\Sigma E(\text{bonds formed})$ when using the bond-energy method?



- A. The bonds in CH_4 and O_2 (reactant bonds)
- B. The bonds in CO_2 and H_2O (product bonds)
- C. All bonds in reactants and products
- D. Only the bonds that change phase ($\text{g} \rightarrow \text{l}$)

11) A student uses average bond energies and gets $\Delta H \approx -780 \text{ kJ}$ for methane combustion, but a data table gives -890 kJ . Which explanation best accounts for the difference?

- A. Bond energies are exact for every molecule, but ΔH° values are rounded.
- B. Bond energies are average values that depend on molecular environment; ΔH° values come from measured reference data.
- C. The sign of ΔH changes when oxygen is a reactant.
- D. Combustion reactions cannot be analyzed using energy ideas.

12) A reaction has $\Delta H < 0$ but still proceeds very slowly at room temperature. Which statement best explains this?

- A. Exothermic reactions must be fast.
- B. The reaction has a large activation energy barrier even though the overall energy change is negative.
- C. The reaction must actually be endothermic.
- D. A catalyst would make ΔH equal to zero.

 Written Response — Diploma-Style Explanation

13) Compare the bond-energy method to the formation-enthalpy (ΔH°_f) method for finding ΔH .

Your response should:

- explain what each method is actually using as evidence/data,
- explain **why** the bond-energy method gives an estimate,
- describe one situation where the bond-energy method is still useful,
- connect your explanation to the ideas of Hess' Law (path independence).

TEACHER VERSION — Full Worked Solutions

Teacher Notes (Using These Solutions)

These solutions are written to model *reasoning*, not just final answers. Where students commonly go wrong, a brief “common error” note is included so you can address misconceptions explicitly.

Solutions — Questions 1-2

Question 1

a) Bonds broken (reactants): $1 \times (\text{H-H})$ and $1 \times (\text{Cl-Cl})$.

Bonds formed (products): $2 \times (\text{H-Cl})$.

b) $\Sigma E(\text{broken}) = 436 + 243 = 679 \text{ kJ}$

$\Sigma E(\text{formed}) = 2(431) = 862 \text{ kJ}$

$\Delta H \approx 679 - 862 = -183 \text{ kJ}$

c) ΔH is negative, so the *system* (the reacting particles) loses energy overall. That energy is transferred to the surroundings as heat, so the surroundings warm up. The reaction is exothermic.

Common error: forgetting to multiply the H-Cl bond energy by 2 (because 2 HCl molecules are formed).

Question 2

a) The “hill” represents the activation energy barrier: particles must collide with enough energy and the right orientation to reach the transition state (where old bonds are stretched/partly broken and new bonds are beginning to form).

b) On an exothermic profile: E_a (forward) is the vertical distance from reactants up to the peak; ΔH is the vertical difference from reactants down to products (negative).

c) If product bonds are stronger overall, more energy is released during bond formation. That lowers the potential energy of the products relative to the reactants, so the product line sits *lower* on the diagram (more negative ΔH).

✓ Solutions — Questions 3–4

Question 3 (Why bond-energy ΔH values are estimates)

Two key reasons:

- 1) Bond energies are *average* values measured across many different molecules. The exact energy to break a bond depends on its molecular environment (neighbouring atoms, resonance, bond angles, etc.).
- 2) The method assumes gas-phase, “average” bonds and does not automatically account for phase changes, intermolecular forces, or specific structural details. A ΔH°_f calculation is based on measured enthalpy changes tied to standard reference states, so it can be more accurate for real conditions.

Question 4 (Catalyst vs energy released)

A catalyst provides an alternate pathway with a lower activation energy, so more collisions have enough energy to react per second. However, ΔH depends only on the energy difference between reactants and products (a state function). Because a catalyst does not change the identities or energies of the initial and final states, ΔH stays the same—only the *rate* changes.

Common error: students say a catalyst “adds energy” or “makes it more exothermic.” It does neither.

✓ Answers — Multiple Choice (Questions 5–12)

- 5) B — You must input energy to overcome the electrostatic attraction that holds the atoms together.
- 6) C — Doubling the reaction doubles every bond count, so ΔH doubles: $-250 \text{ kJ} \rightarrow -500 \text{ kJ}$.
- 7) B — Exothermic means energy released on bond formation exceeds energy absorbed on bond breaking.
- 8) C — A catalyst lowers E_a but does not change ΔH .
- 9) B — Breaking weak and forming strong bonds favours a negative ΔH (more energy released).
- 10) B — “Bonds formed” refers to bonds in products.

11) B — Average bond energies vary by environment; ΔH° values are tied to standard reference data and conditions.

12) B — A large E_a can make a reaction slow even if it is energetically favourable ($\Delta H < 0$).

Diploma Tip — What a Strong Written Response Looks Like (Question 13)

Top-level responses typically do all of the following:

- Clearly define each method:
 - Bond-energy method: uses average bond energies and counts bonds broken/formed to *estimate* ΔH .
 - Formation-enthalpy method: uses ΔH°_f values and the relationship $\Delta H^\circ_{\text{rxn}} = \sum n\Delta H^\circ_f(\text{products}) - \sum n\Delta H^\circ_f(\text{reactants})$.
- Explain *why* bond-energy values are approximate (average, environment-dependent, phase/intermolecular effects).
- Give a sensible use-case: e.g., quick prediction of sign/magnitude, comparing pathways, or checking whether an answer is reasonable when ΔH°_f data are missing.
- Connect to Hess' Law: both rely on the idea that overall ΔH depends only on initial and final states (path independent), so we can “re-route” the calculation using different data sources.

Model Response — Question 13 (Sample)

The bond-energy method estimates ΔH by counting the bonds that must be broken in the reactants and the bonds that form in the products. Because breaking bonds absorbs energy and forming bonds releases energy, we use $\Delta H \approx \sum E(\text{broken}) - \sum E(\text{formed})$ to decide whether the reaction is overall exothermic or endothermic. This method gives an estimate because bond energies are average values: the exact energy of a “C–H” or “C=O” bond depends on the molecule’s structure and chemical environment. In addition, bond-energy calculations typically assume average gas-phase bonds and do not fully capture details such as intermolecular forces and phase changes, so small differences from experimental ΔH values are expected.

In contrast, the formation-enthalpy method uses ΔH°_f values that are measured relative to standard reference states. We calculate $\Delta H^\circ_{\text{rxn}} = \sum n\Delta H^\circ_f(\text{products}) - \sum n\Delta H^\circ_f(\text{reactants})$. Under standard conditions, this approach is typically more accurate

because the ΔH°_f data come from experimental enthalpy measurements tied to real substances in their standard states.

Even though it is less precise, the bond-energy method is still useful when ΔH°_f values are not available, or when you want a quick, evidence-based prediction of sign and approximate size as a reasonableness check. Both approaches connect to Hess' Law because they rely on path independence: the total enthalpy change depends only on the starting and ending states, so we can choose a convenient route (bond energies or formation enthalpies) to calculate the same overall ΔH .

✓ Solution 6

a) Bonds broken: H—H (436) + Cl—Cl (242) = 678 kJ/mol.

b) Bonds formed: $2 \times \text{H—Cl} = 2 \times 431 = 862$ kJ/mol (released).

c) $\Delta H \approx \Sigma E(\text{broken}) - \Sigma E(\text{formed}) = 678 - 862 = -184$ kJ/mol. Negative means exothermic.

d) Why this works: breaking bonds requires energy input to pull atoms apart; forming bonds releases energy because the product bond is more stable (lower potential energy). The overall sign depends on whether bond-forming release outweighs bond-breaking cost.

✓ Solution 7

a) E_a is a property of the reaction pathway and does not change just because temperature changes.

b) Higher temperature means particles have higher average kinetic energy and a wider spread of energies. More particles have $E \geq E_a$, so more collisions are successful per second.

c) On an energy diagram, the curve (including E_a and ΔH) stays the same. Temperature changes how many particles can 'get over' the barrier, not the barrier itself.

✓ Solution 8

a) Gasoline combustion is exothermic (large negative ΔH), but it has a significant activation energy. At room temperature, most collisions between gasoline vapour and

oxygen do not have enough energy to reach the activated complex.

b) A spark/flame supplies the initial energy needed to overcome the activation barrier—once started, the reaction can sustain itself.

c) In energy systems, controlling activation energy helps balance safety (storage stability) and usefulness (rapid energy release when desired).

✓ Solution 9–12

9) B. Products lower than reactants means $\Delta H < 0$, exothermic.

10) C. Use $E_a(\text{rev}) = E_a(\text{fwd}) - \Delta H = 60 - (-40) = 100 \text{ kJ}$.

11) C. A catalyst provides an alternative pathway with lower E_a .

12) C. $E_a(\text{fwd})$ is measured from reactants up to the peak.

✓ Solution 13

A strong explanation includes these points:

- Bond breaking always requires energy input. Early in the reaction, reactant bonds must be stretched and partially broken before new bonds can form.
- That bond-breaking 'investment' is the activation energy.
- Once the activated complex is reached, new bonds form and release energy. If the release is greater than the initial cost, the overall ΔH is negative (exothermic).
- So: ΔH describes the start-to-finish change, while E_a describes the barrier you must cross to get the reaction started.

✓ Solution 14

Rate-determining step:

- On a multi-step energy profile, the rate-determining step is the step with the largest activation energy barrier (largest peak relative to the previous valley).
- That step is slowest, so it controls the overall rate.

Overall ΔH :

- ΔH depends only on initial and final energies (reactants vs products). Intermediate steps and peaks change the pathway but do not change the start-to-finish energy difference.
- This is the same 'path-independence' idea you will use again with Hess's Law.

✓ Solution 15

A strong STS response should include:

- A named example (enzymes / catalytic converter / Haber process).
- A statement that catalysts lower E_a by offering an alternative pathway.
- A link to practical outcomes: faster reactions at lower temperature, less energy input, better control, improved safety, reduced emissions or waste.

Example (catalytic converter): It uses metal catalysts to lower E_a for oxidation/reduction reactions that convert CO and NO_x into less harmful gases. Lower E_a means the reactions proceed quickly at exhaust temperatures, improving air quality.

LESSON 6 SELF-CHECK (before you move on)

- ✓ I can explain activation energy as an energy barrier and describe what the peak means on an energy diagram.
- ✓ I can calculate $E_a(\text{fwd})$, $E_a(\text{rev})$, and ΔH when energy values are provided.
- ✓ I can distinguish clearly between E_a (barrier height) and ΔH (start-to-finish energy change).
- ✓ I can explain catalysts as alternative pathways that lower E_a without changing ΔH .
- ✓ I can justify answers using diagram language (reactants, products, activated complex, pathway).

🔗 Chemistry 30 | Thermochemical Changes | Lesson 7

Bond Energy and Reactions

Student Practice Pack (Diploma-Style) • questions only

🌱 Big Idea

Bond energies help us estimate reaction enthalpy by comparing the energy absorbed to break bonds with the energy released when new bonds form.

✓ How to Use This Practice Pack

This set is designed to feel like diploma questions: you will need to interpret information, connect ideas, and justify your reasoning.

Show units, keep track of signs (energy absorbed vs released), and write a short sentence that explains what your final ΔH means physically.

If a bond energy table is provided, use those values exactly (do not substitute memory values).

📄 Data: Average Bond Energies (kJ/mol)

Use these average bond energies unless a question provides different values:

H-H 436 | O=O 498 | O-H 463

C-H 413 | C-C 347 | C=C 614

C-O 358 | C=O (in CO₂) 799 | C=O (carbonyl) 743

N-H 391 | N≡N 945 | C-Cl 338

Units: kJ/mol (per mole of bonds). These are averages → your ΔH values are estimates.

🔍 Section A — Bond-Energy Calculations (Numerical Response)

For each problem: (1) count bonds broken and formed, (2) calculate $\Sigma E(\text{broken})$ and $\Sigma E(\text{formed})$, (3) compute $\Delta H \approx \Sigma E(\text{broken}) - \Sigma E(\text{formed})$, and (4) interpret the sign.

1) Combustion of hydrogen: $2 \text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2 \text{H}_2\text{O}(\text{g})$

a) Estimate ΔH for the reaction using the bond energies provided above.

b) Explain, in one sentence, why this reaction warms the surroundings.

- 2) Formation of ammonia: $\text{N}_2(\text{g}) + 3 \text{H}_2(\text{g}) \rightarrow 2 \text{NH}_3(\text{g})$
- Estimate ΔH using bond energies.
 - The bond-energy method uses averages. Identify one reason your value might differ from a tabulated ΔH° value.
- 3) Chlorination (simplified): $\text{CH}_4(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow \text{CH}_3\text{Cl}(\text{g}) + \text{HCl}(\text{g})$
Given: Cl-Cl 242 kJ/mol, H-Cl 431 kJ/mol.
- Estimate ΔH .
 - Based on your sign, is the reaction exothermic or endothermic?
- 4) Hydrogenation: $\text{C}_2\text{H}_4(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{C}_2\text{H}_6(\text{g})$
- Estimate ΔH .
 - Connect your result to what happens to the number/type of bonds in the products.
- 5) A reaction has $\Delta H \approx -95 \text{ kJ}$ (from bond energies). State whether it is exothermic or endothermic and describe the energy flow using 'system' and 'surroundings'.
- 6) Two students use the same bond energies but one gets a positive ΔH and the other gets a negative ΔH .
List two specific calculation mistakes that commonly cause the sign to flip.



Section B — Energy Diagrams & Reaction Profiles (Interpretation)

Answer using complete sentences where asked. Use diagram language precisely: reactants, products, activation energy (E_a), activated complex, and ΔH .

- 7) Sketch (or describe clearly) an exothermic reaction profile and label: reactants, products, ΔH , and E_a .
Then explain how you can tell it is exothermic from the diagram.
- 8) A reaction profile shows products 40 kJ above reactants, and the peak is 120 kJ above reactants.
- Determine ΔH .
 - Determine E_a (forward) and E_a (reverse).
 - State whether the forward reaction is endothermic or exothermic.
- 9) A catalyst is added to an exothermic reaction.
- Describe what changes on the energy diagram.

- b) State two things that do NOT change.
- c) Explain why the overall ΔH stays the same even though the pathway changes.

10) A student says: "If E_a is large, the reaction must be endothermic."

Explain why this is incorrect, using a reaction profile argument.

 Common Error

Mixing up ΔH and E_a :

- ΔH compares products to reactants (vertical difference between start and end).
- E_a compares reactants to the peak (vertical difference between start and the highest point).

A reaction can be exothermic ($\Delta H < 0$) and still have a large activation energy.

 Section C — Diploma-Style Written Response (Section B/C)

These questions assess reasoning. Use clear, evidence-based explanations.

11) Bond-energy method as an estimate:

Explain why ΔH values calculated using average bond energies are not exactly the same as experimental ΔH° values. In your explanation, include at least two distinct reasons and connect each reason to the idea of 'average' bond energies.

12) Evidence-based exothermic/endothermic judgement:

You calculate $\Delta H \approx +65 \text{ kJ}$ for a reaction using bond energies. Another student argues the reaction is still likely exothermic 'because it happens easily'.

- a) Explain what your calculated ΔH implies about energy flow.
- b) Explain why 'happens easily' is a statement about E_a /rate, not ΔH .
- c) State one type of evidence you would look for to decide whether the reaction is fast or slow.

Pause & Think (Bridge)

Bond energies tell us about the energy bookkeeping (ΔH). Activation energy tells us about the energy barrier (E_a).

In the next lesson, you will use these ideas to explain why changing temperature (or adding a catalyst) affects reaction rate—and why that is a different question than whether the reaction releases energy.

Chemistry 30 | Thermochemical Changes | Lesson 7

Bond Energy and Reactions

Worked solutions and rationales

Big Idea

In bond-energy problems, ΔH is estimated by comparing energy absorbed to break bonds with energy released when new bonds form: $\Delta H \approx \Sigma E(\text{broken}) - \Sigma E(\text{formed})$. Reaction profiles then separate ΔH (energy change) from E_a (barrier).

How to Mark / What to Look For

For numerical response: students should show bond counts, substitutions with units, and a final statement interpreting the sign.

For diagrams: labels must be placed correctly (E_a from reactants to peak; ΔH from reactants to products).

For written response: look for clear separation of 'energy change' (ΔH) vs 'rate/barrier' (E_a), and for accurate use of system/surroundings language.

 Data (same values as student pack)

Reference bond energies used in solutions (kJ/mol):

H-H 436; O=O 498; O-H 463; C-H 413; C-C 347; C=C 614;

C-O 358; C=O (CO₂) 799; C=O (carbonyl) 743; N-H 391; N≡N 945; C-Cl 338;

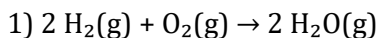
Cl-Cl 242; H-Cl 431.

 Section A — Worked Numerical Solutions

Method reminder: $\Delta H \approx \Sigma E(\text{bonds broken}) - \Sigma E(\text{bonds formed})$.

If ΔH is negative → exothermic (system releases energy to surroundings).

If ΔH is positive → endothermic (system absorbs energy from surroundings).



Bonds broken:

• $2 \times (\text{H-H}) = 2(436) = 872 \text{ kJ}$

• $1 \times (\text{O=O}) = 498 \text{ kJ}$

$\Sigma E(\text{broken}) = 872 + 498 = 1370 \text{ kJ}$

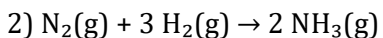
Bonds formed:

• $4 \times (\text{O-H}) = 4(463) = 1852 \text{ kJ}$

$\Sigma E(\text{formed}) = 1852 \text{ kJ}$

$\Delta H \approx 1370 - 1852 = -482 \text{ kJ}$

Interpretation: Exothermic. The system releases ~482 kJ per reaction as written, so the surroundings warm.



Broken:

• $1 \times (\text{N}\equiv\text{N}) = 945 \text{ kJ}$

• $3 \times (\text{H-H}) = 3(436) = 1308 \text{ kJ}$

$\Sigma E(\text{broken}) = 945 + 1308 = 2253 \text{ kJ}$

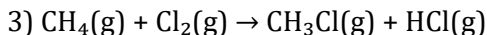
Formed:

- In 2 NH₃ there are 6 N-H bonds → 6(391) = 2346 kJ

$$\Delta H \approx 2253 - 2346 = -93 \text{ kJ}$$

Interpretation: Exothermic by estimate.

Why might it differ from ΔH° tables? (Any one is acceptable) bond energies are averages; real bonds differ by molecular environment; phase/temperature differences; experimental ΔH° includes all energy changes under stated conditions.



Think: one C-H bond is replaced by one C-Cl bond, and Cl₂ provides the Cl while forming H-Cl.

Broken:

- 1 × (C-H) = 413 kJ

- 1 × (Cl-Cl) = 242 kJ

$$\Sigma E(\text{broken}) = 655 \text{ kJ}$$

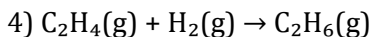
Formed:

- 1 × (C-Cl) = 338 kJ

- 1 × (H-Cl) = 431 kJ

$$\Sigma E(\text{formed}) = 769 \text{ kJ}$$

$$\Delta H \approx 655 - 769 = -114 \text{ kJ} \rightarrow \text{exothermic.}$$



Broken:

- 1 × (C=C) = 614 kJ

- 1 × (H-H) = 436 kJ

$$\Sigma E(\text{broken}) = 1050 \text{ kJ}$$

Formed:

- 1 × (C-C) = 347 kJ

- 2 × (C-H) = 2(413) = 826 kJ

$$\Sigma E(\text{formed}) = 347 + 826 = 1173 \text{ kJ}$$

$$\Delta H \approx 1050 - 1173 = -123 \text{ kJ} \rightarrow \text{exothermic.}$$

Bond-type connection: a π -bond (part of C=C) is replaced by σ -bonds (C-C plus two C-H), and forming stronger/more stable bonds releases energy.

5) If $\Delta H \approx -95 \text{ kJ}$:

Negative ΔH means the system loses enthalpy. Energy flows out of the system into the surroundings (often as heat), so the surroundings warm.

6) Two common mistakes that flip the sign (any two):

- Swapping the order and calculating $\Sigma E(\text{formed}) - \Sigma E(\text{broken})$ instead of broken – formed.
- Miscounting bonds (especially double bonds vs single bonds, or forgetting that coefficients multiply the entire molecule's bonds).
- Using the wrong bond type (C=O in CO₂ vs carbonyl, etc.).
- Treating a double bond as 'two single bonds' without using the correct bond energy.



Section B — Energy Diagram Solutions and Marking Notes

Key formulas/relationships:

$\Delta H = E(\text{products}) - E(\text{reactants})$ (vertical difference end – start).

$E_a(\text{forward}) = E(\text{peak}) - E(\text{reactants})$.

$E_a(\text{reverse}) = E(\text{peak}) - E(\text{products})$.

7) Exothermic profile (expected features):

- Products lower than reactants.
- ΔH arrow points downward from reactants level to products level ($\Delta H < 0$).
- E_a arrow points upward from reactants level to the peak.

Explanation: exothermic because final potential energy (products) is lower than initial (reactants), so energy is released to surroundings.

8) Given: products are 40 kJ ABOVE reactants; peak is 120 kJ ABOVE reactants.

a) $\Delta H = +40$ kJ (endothermic, system gains energy).

b) $E_a(\text{forward}) = 120$ kJ.

$E_a(\text{reverse}) = 120 - 40 = 80$ kJ (because products start closer to the peak).

c) Forward reaction is endothermic ($\Delta H > 0$).

9) Catalyst on an exothermic reaction:

a) Energy diagram change: peak becomes lower (smaller E_a). Sometimes shown as a new pathway with a lower maximum.

b) Does NOT change: reactant energy level, product energy level, and therefore ΔH . Also does not change the equilibrium enthalpy difference.

c) Why ΔH stays the same: ΔH is a state function—depends only on initial and final states, not the pathway. A catalyst changes pathway (rate), not overall energy bookkeeping.

10) Why 'large E_a means endothermic' is incorrect:

E_a measures the barrier height, not the start-to-finish difference.

You can have products lower than reactants ($\Delta H < 0$, exothermic) but still have a very high peak (large E_a). That reaction releases energy overall but may be slow to start.

 Section C — Model Written Responses (with key points)

Use these as exemplars. Student wording may vary, but the scientific ideas must be correct.

11) Why bond-energy ΔH values are estimates (model response):

Bond energies in tables are average values measured across many different molecules. A C–H bond in methane does not have exactly the same energy as a C–H bond in an alcohol, for example, because the surrounding atoms change electron distribution and bond strength. When we use one average number for every C–H bond, we build approximation into the calculation.

Also, the bond-energy method treats the reaction as if bonds are broken and formed in the gas phase with no other energy changes. Real ΔH° values can include phase effects (g vs l vs s) and other interactions. Because the table values are averages and conditions vary, the result is best interpreted as a reasonable estimate of both sign and approximate magnitude.

12) Interpreting ΔH and separating it from 'happens easily' (model response):

a) A positive ΔH means the system absorbs energy overall. Energy must flow from the surroundings into the reacting system; the surroundings would tend to cool unless energy is supplied.

b) 'Happens easily' describes how quickly the reaction can begin, which depends on activation energy and collision conditions. A reaction can be endothermic ($\Delta H > 0$) and still happen quickly if E_a is small, or exothermic ($\Delta H < 0$) and happen slowly if E_a is large.

c) Evidence for fast/slow: compare reaction rates at different temperatures, look for need for heating/spark, measure time to completion, or compare catalyzed vs uncatalyzed rates—these relate to E_a and kinetics, not to ΔH .

 Quick Answer Key (at a glance)

Q1 -482 kJ (exo).

Q2 -93 kJ (exo, estimate).

Q3 -114 kJ (exo).

Q4 -123 kJ (exo).

Q8 $\Delta H +40 \text{ kJ}$; $E_a(\text{fwd}) 120 \text{ kJ}$; $E_a(\text{rev}) 80 \text{ kJ}$.

Catalyst lowers E_a only; ΔH unchanged.

 Chemistry 30 | Thermochemical Changes | Lesson 7

Bond Energy and Reactions

Exit Ticket • 1–2 diploma-style prompts (bridge to next lesson)

 Big Idea

ΔH answers: “Is energy released or absorbed overall?” E_a answers: “How hard is it to start?”
Bond energies help you reason about both—if you use them carefully.

 Instructions

Answer in complete sentences. Use key terms correctly (system, surroundings, ΔH , E_a , catalyst, bond breaking/forming).

You are not being marked on artistic quality—if you sketch a reaction profile, labels are what matter.

 Exit Prompt 1 (Written Response)

A student says: "If a reaction releases a lot of energy, it must be fast."

Respond to the student by clearly separating ΔH from E_a . Include one example of how a reaction could have a very negative ΔH but still be slow.

 Exit Prompt 2 (Bridge to Kinetics/Equilibrium)

Consider an exothermic reaction that reaches equilibrium in a closed container.

- a) Sketch or describe how adding a catalyst would change the reaction profile.
- b) Explain what changes about the time needed to reach equilibrium.
- c) Explain what does NOT change about the equilibrium position, and why.

 Looking Ahead

Next lesson, you will connect energy ideas to reaction rate and to equilibrium decisions.

A useful habit: whenever you see "fast/slow," think kinetics and E_a ; whenever you see "more products at equilibrium," think equilibrium and Le Châtelier; whenever you see "heat released/absorbed," think ΔH .